Smart Meters – Smart Behaviors? An Analysis of London Household Energy Use in Response to Dynamic Time-of-Use Tariffs

ENV 797: Time Series Analysis Final Project

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The GitHub repository for this projects is located here: https://github.com/sylviahipp/GualtieriReedHippKastenSalzberg_ENV797_TSA_FinalProject

Introduction

The mass adoption of smart meters has enabled the next generation of grid management, through bidirectional, real-time communication of electricity use. This is providing both end users and grid managers with high resolution data that can enhance power optimization and management, which results in real energy and cost savings (Gholamnia et al., 2025, p. 5238; Meeks et al., 2024, p. 2; The Benefits of Smart Meters, n.d.; Walsh, 2025). Making use of this enhanced data resolution, utilities and grid operators are exploring the relevance of implementing time-variable pricing schemes such as Time of Use (ToU) tariffs, which better account for the true marginal costs of electricity ("Demand Response and Time-Variable Pricing Programs", 2020). Static ToU tariffs impose different prices on electricity use according to market demand, commonly in 3 tiers such as off-peak, mid-peak or normal, and peak (Enrich et al., 2024, p. 2). Some programs integrate dynamic pricing (dToU), which can increase or decrease peak prices on various time-scales (i.e. at daily or near-real time intervals) to further induce demand response from customers (Gholamnia et al., 2025, pp. 5239–5240).

The literature generally considers demand response as the central goal of implementing ToU tariffs; mitigating extreme peak loads avoids requiring the highest marginal cost (and often most heavily polluting) plants to come online (Barjaková et al., 2024, p. 3; Enrich et al., 2024, p. 3; Meeks et al., 2023, pgs. 12-13; "What Are the Benefits of Time-Of-Use Pricing Policies?", 2025). While the environmental benefits may be significant, consumers generally see this as a pleasant co-benefit to the significantly more compelling opportunity to save money on energy bills (Barjaková et al., 2024, p. 9).

According to a systematic literature review by Yuniarto et al. (2023), the effectiveness of ToU tariffs has been increasingly studied, particularly from the user behavior perspective. The behavioral dimensions of ToU was also studied by Barjaková et al. (2024), in an effort to understand which educational outreach materials were most effective, and the true motivations of enrollees. These behavioral studies add rich nuance to the longstanding quantitative analyses of the impacts and successes of ToU tariffs. For example, Faruqui and Sergici conducted a meta-analysis of 34 studies a decade prior, which used regression analysis to assess the enabling capacity of smart meters and other technologies on ToU effectiveness (2023, p. 58). Through their study, they identified two "arcs of price responsiveness," which revealed that smart technology supported tariffs were notably more effective at driving peak reductions; for example, they note that given a 5:1 ratio of peak price to off-peak price, smart technology enabled tariffs would expect a nearly 9% greater reduction in peak load, achieving a roughly 20% net reduction (ibid., p. 61). Other individual studies and meta-analyses have found peak and mid-peak reductions on the order of ~5% to 15% (Enrich et al., 2024; Fu et al., 2024, p 3; Harding & Lamarche, 2016, p. 908).

Crucially, these past studies have relied on various forms of advanced regression analysis to compare the effects of ToU between groups. The use of time series analysis has, so far, been limited to designing dToU tariffs or comparing critical peak pricing (a different form of time-varying pricing) and ToU (Gholamnia et al., 2025; Schittekatte et al., 2024). Our project contributes to the growing literature on Time of Use Tariffs by using time series analysis to assess whether dynamic ToU tariffs result in demand response compared to modeled counterfactual baselines, thus comparing residential energy usage within the same groups.

Research Question and Hypothesis

Does demand forecasted under pre-ToU conditions differ from actual demand once ToU tariffs are introduced?

We expect that customers will change their electricity usage behavior based on ToU tariff rates and shift some demand from traditional peak periods when tariff rates are high to periods when tariff rates are low. We hypothesize that the forecasted versus actual for customers with a standard rate would show normally distributed residuals around 0 while the ToU customers have skewed residuals, based on the tariff rate level. During high tariff hours, we expect that the forecasted value is higher than the actual value and that during low tariff hours, customers actually use more demand than forecasted.

Dataset Information

Raw Data

The dataset for this project was obtained from a study done by UK Power Networks titled SmartMeter Energy Consumption Data in London Households.[1] The dataset includes energy consumption readings every half hour over the period November 2011 to February 2014. The purpose of the study was to understand how price signals could support higher renewable penetration as well as reduce stress on local distribution grids. 5,566 households participated in the study and were selected as a balanced sample to represent the electricity usage of the Greater London population.

1,123 of the participating households were subject to Dynamic Time of Use (dToU) energy prices throughout 2013. The prices were given to the households a day ahead and were displayed on the customers' smart meter or texted to them directly. Tariffs ranged from High (67.20p/kWh), Normal (11.76p/kWh), or Low (3.99p/kWh). The remaining 4,443 customers ("Standard" households) were given a flat rate tariff of 14.228p/kWh throughout the entire study period. A glimpse of the raw load data by household is shown in [[**Table 1**]] below, where `LCLid` is the unique household i.d., and `stdorTou` indicates whether the households were given standard tariff rates or subject to dToU tariffs. A glimpse of the raw data containing the tariff schedule throughout 2013 is shown in [[**Table 2**]].

Table 1: Household Load Raw Data

LCLid	stdorToU	DateTime	kWhPerHalfHour
MAC000002	Std	2013-01-01 00:00:00.0000000	0.219
MAC000002	Std	2013-01-01 00:30:00.0000000	0.241
MAC000002	Std	2013-01-01 01:00:00.0000000	0.191
MAC000002	Std	2013-01-01 01:30:00.0000000	0.235
MAC000002	Std	2013-01-01 02:00:00.0000000	0.182

Table 2: Tariff Schedule Raw Data

TariffDateTime	Tariff
2013-01-01 00:00:00	Normal
2013-01-01 00:30:00	Normal
2013-01-01 01:00:00	Normal
2013-01-01 01:30:00	Normal
2013-01-01 02:00:00	Normal

Data Wrangling

Both the household load and tariff schedule data were aggregated to the hour as the native half-hour increments complicated the time series seasonality without improving the model. Household load was given as kWh per half hour for each household, so hourly load was taken as the total load over both half hours.[2] For the tariff schedule data, in cases where the tariff level differed between the top and bottom of the hour, the following logic was applied to infer tariff level for the whole hour:

1) High and Normal –> High
2) Low and Normal –> Low

[1] UK Power Networks. (2015). *SmartMeter energy consumption data in London households.* London Datastore. https://data.london.gov.uk/dataset/smartmeter-energy-consumption-data-in-london-households-vqm0d/.

[2] The initial dataset of household load included over 200 million observations representing 2+ years of data at half-hour increments for over 5,500 households. To make the data more manageable in R, the data was initially wrangled in Python then exported to R. The code for this initial data wrangling can be found in `~/scripts/build/00_lcl_data_build.ipynb`.

3) High and Low –> Normal

After aggregating load data to the half hour, the load dataset was processed to exclude missing data. First, the entire dataset was limited to the period January 2012–December 2013 to exclude periods with sparse data in the last two months of 2011 and first two months of 2014. Second, households with over 2,880 hours of missing data (roughly four months) in 2012 were excluded from the analysis.[3] Finally, any household exhibiting extreme outlier behavior was removed.[4] Following this process, 1,528 Standard households and 362 dToU households remained.

Once the final sample of households was identified, households were separated by treatment group (Standard vs. dToU) and k-means clustering was used to aggregate households based on consumption behavior. Only load data from 2012 was used to identify clusters so to avoid capturing any changes in behavior caused by the treatment. First, each household's hourly demand was normalized relative to its total demand in 2012 so that clusters were identified based on energy usage patterns rather than total volume. For both the Standard and dToU households, the optimal number of clusters (i.e., clustering with the maximum silhouette score) was found to be 2. Each treatment group/cluster was then aggregated by taking the mean load at each hour across all households in the group. This resulted in four distinct hourly time series depicted below in [[**Figure 1**]][5] and summarized in [[**Table 3**]]**.** As is shown in the figure, cluster 1 for both the standard and dToU groups exhibited highly varied usage between the winter and summer, while cluster 2 of both groups exhibited much smoother seasonality.

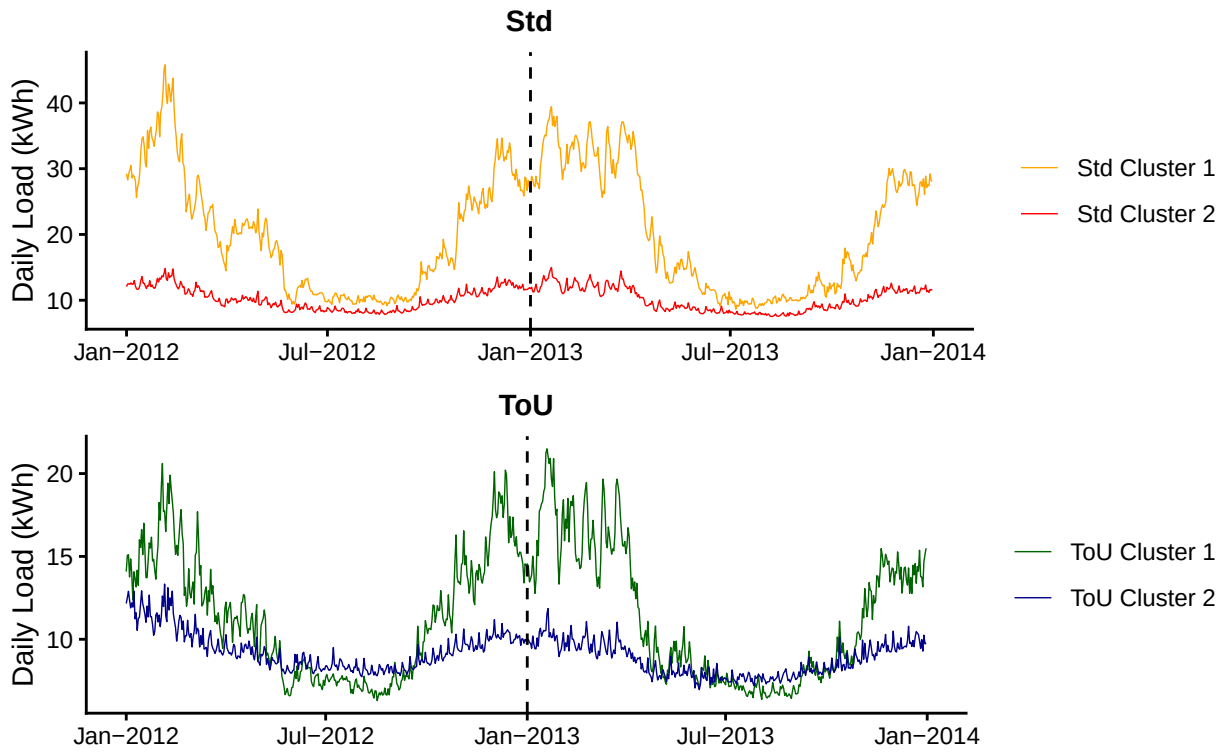

Figure 1: Daily Average Energy Usage By Cluster and Treatment Group

[3]The models in this project were trained on household load data from 2012. Therefore, households with excessive missing data in this year were excluded.

[4]A household was identified as an outlier if the cumulative distance of its hourly usage from the mean was in the 99th percentile among all households.

[5][[**Figure 1**]] shows load data aggregated at the daily level, but the time series used in the modeling and analysis are at the hourly level.

Table 3: Summary of Household Clusters

Treatment Group	Cluster	Number of Households	Max Hourly Load (kW)	Average Hourly Load (kW)
Standard	Cluster 1	49	7.1062917	0.8426666
Standard	Cluster 2	1479	0.9647529	0.4221099
dToU	Cluster 1	104	1.6109524	0.4740610
dToU	Cluster 2	258	1.0514250	0.3776928

Analysis

Exploratory Analysis

All clusters hourly time series show a similar pattern of peaking in the winter and dipping in the summer [[**FIGURE 2**]]. Std1 shows much higher variability and total load than any of the other three clusters, while ToU2 cluster shows the least variability between peak and not peak months. A visual analysis indicates that this data has multiple seasonal components, shown by the peaking and troughing within shorter and longer time periods.

Because a model will be trained on the 2012 data, each cluster was decomposed for 2012 only. The decomposition, as shown in [[**FIGURE 3**]], verifies that there are strong daily and weekly seasonal components. Because this data is only 1 year, the trend does not show an overall increase or decrease in load but likely captures the monthly component.

The ACF and PACF of the residuals of the decomposition show that there is seasonal dependence that is not accurately accounted for in the decomposition ([[**FIGURE 4-7**]]). The ACF shows a slow decay, indicating a persisting strong autoregressive component. Additionally, the ACF and PACF both show peaks at the 24 hour marks, indicating that the hourly component is not fully modeled in the decomposition. Therefore, complex models will be needed to accurately model all seasonal components.

[[**FIGURE 8**]] shows that at the hourly level, tariff rates in 2013 tended to be highest during the hours 17 - 22.

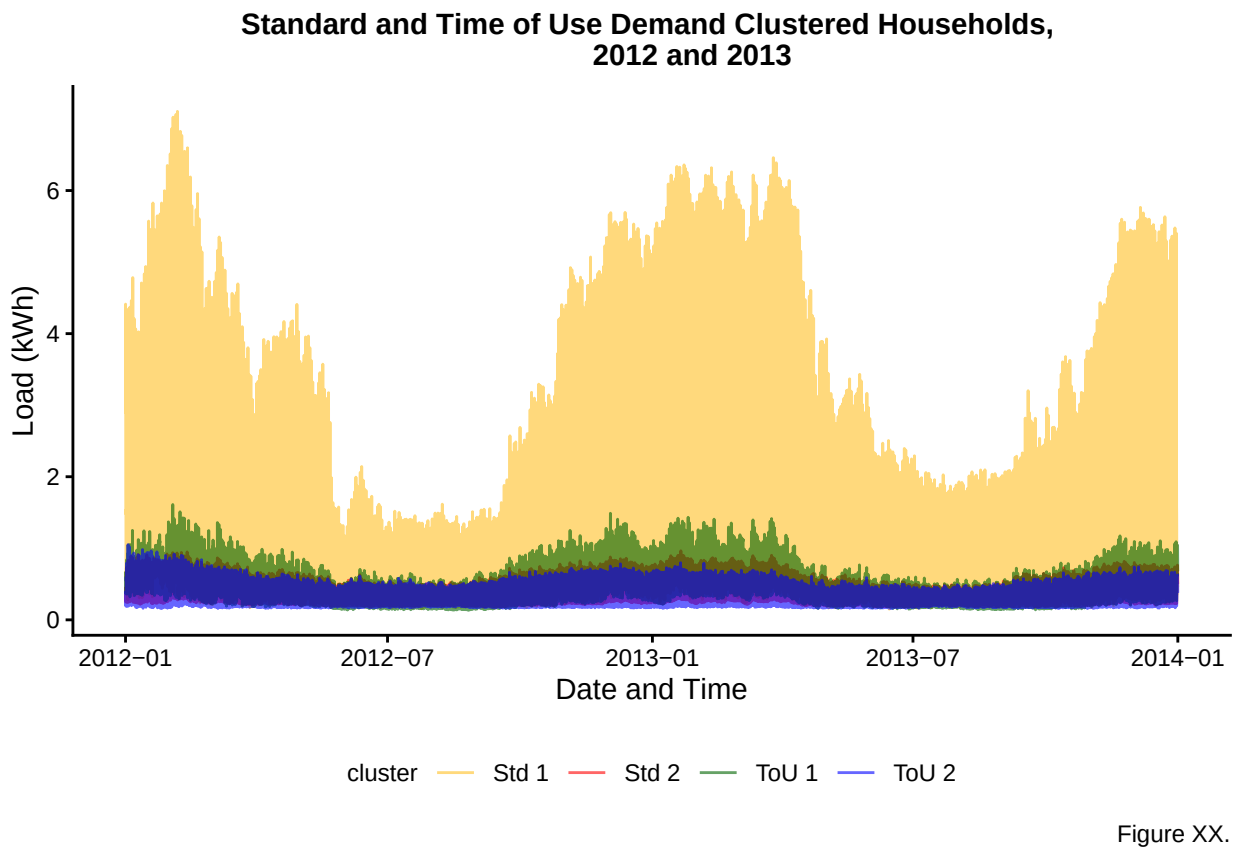

Figure 2: Standard and Time of Use Demand Clustered Households, Hourly Time Series, 2012 and 2013

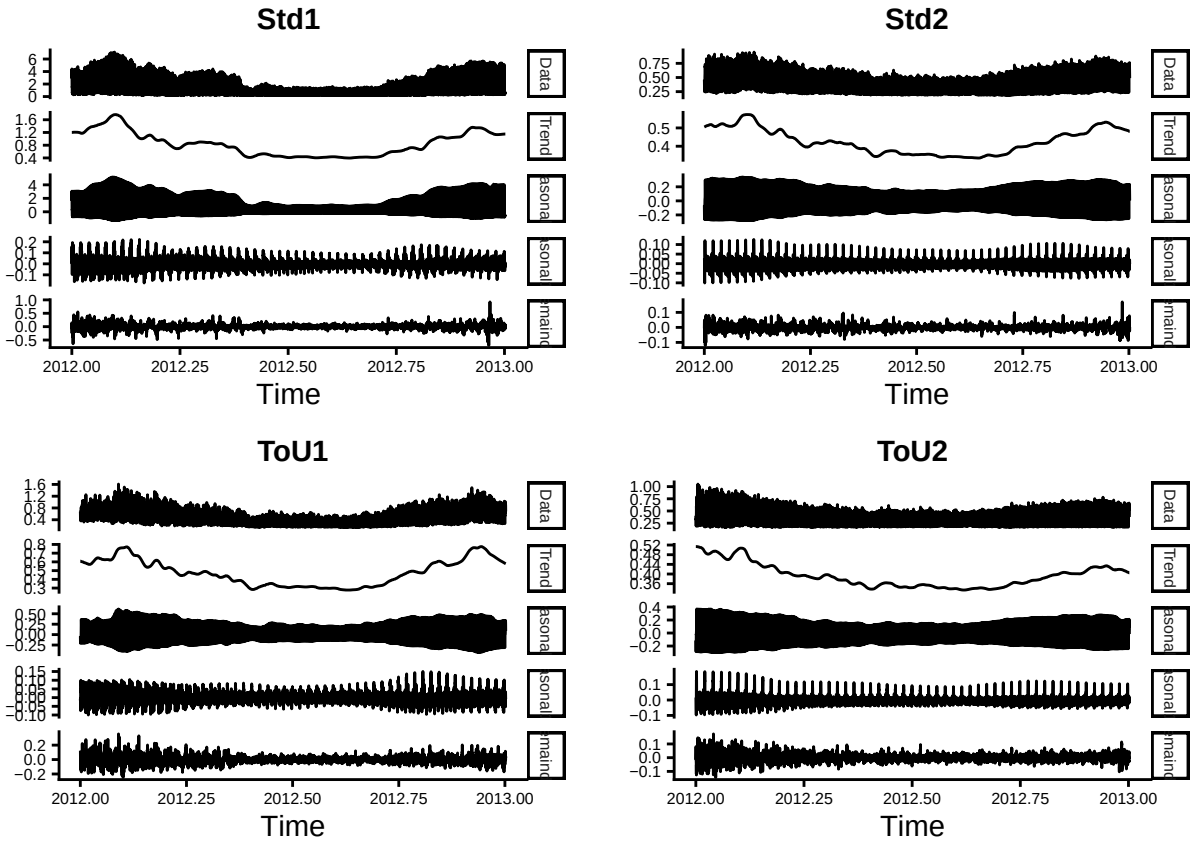

Figure 3: Figure XX. Decomposition of each clusters, 2012

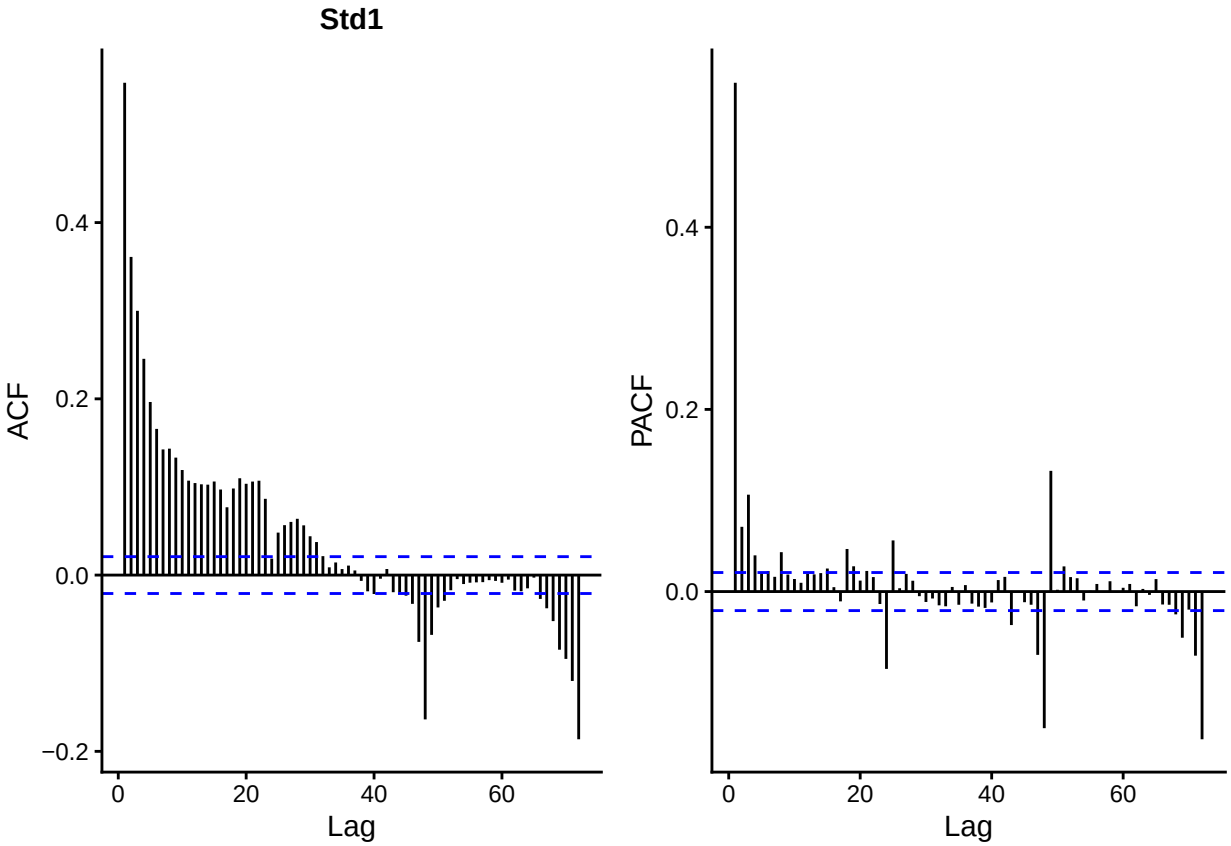

Figure 4: Std1 ACF and PACF

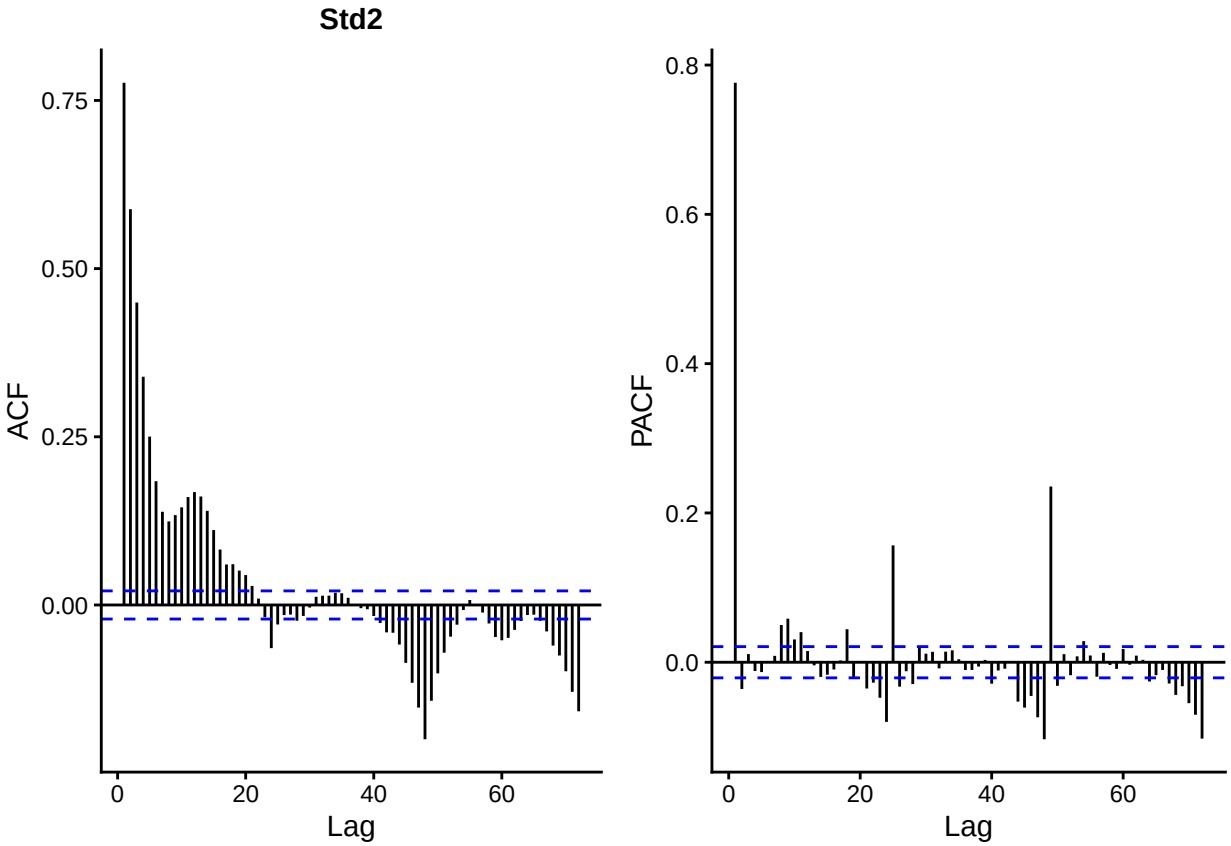

Figure 5: Std2 ACF and PACF

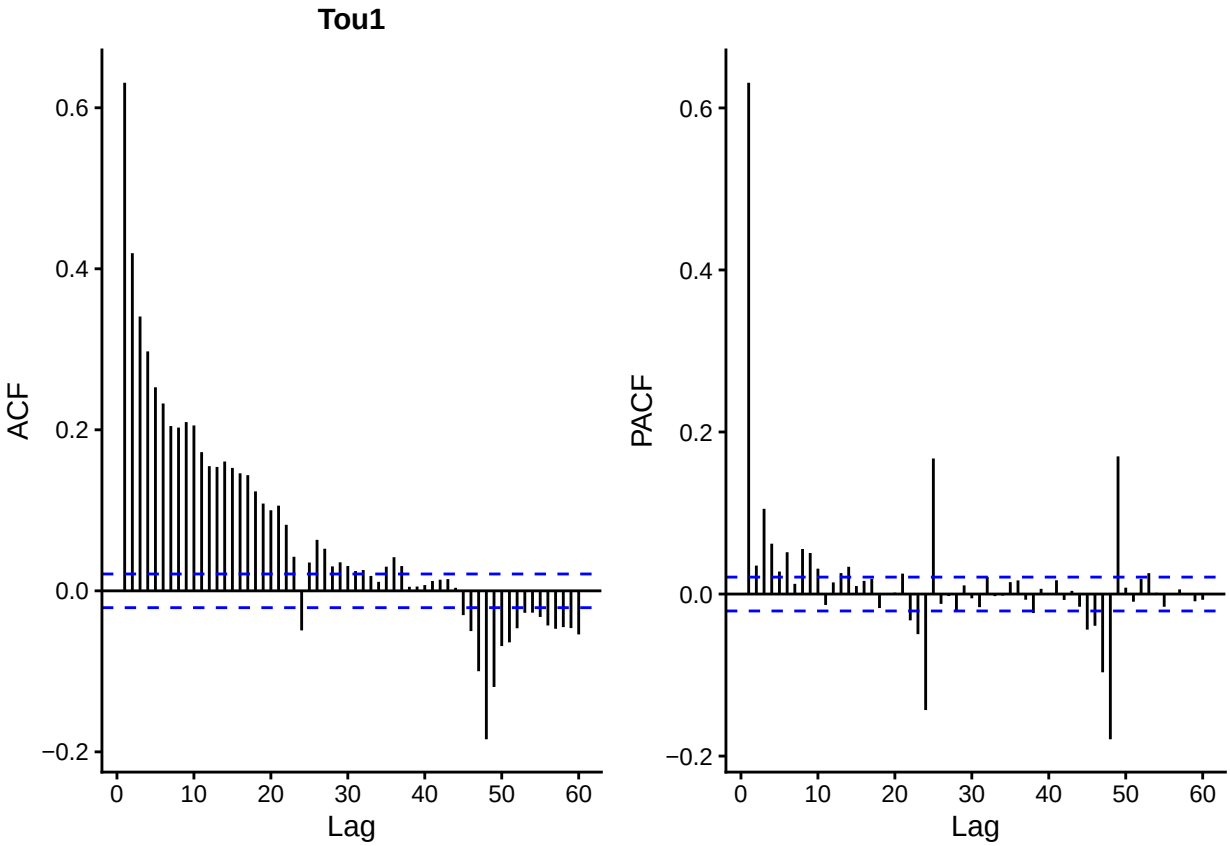

Figure 6: ToU1 ACF and PACF

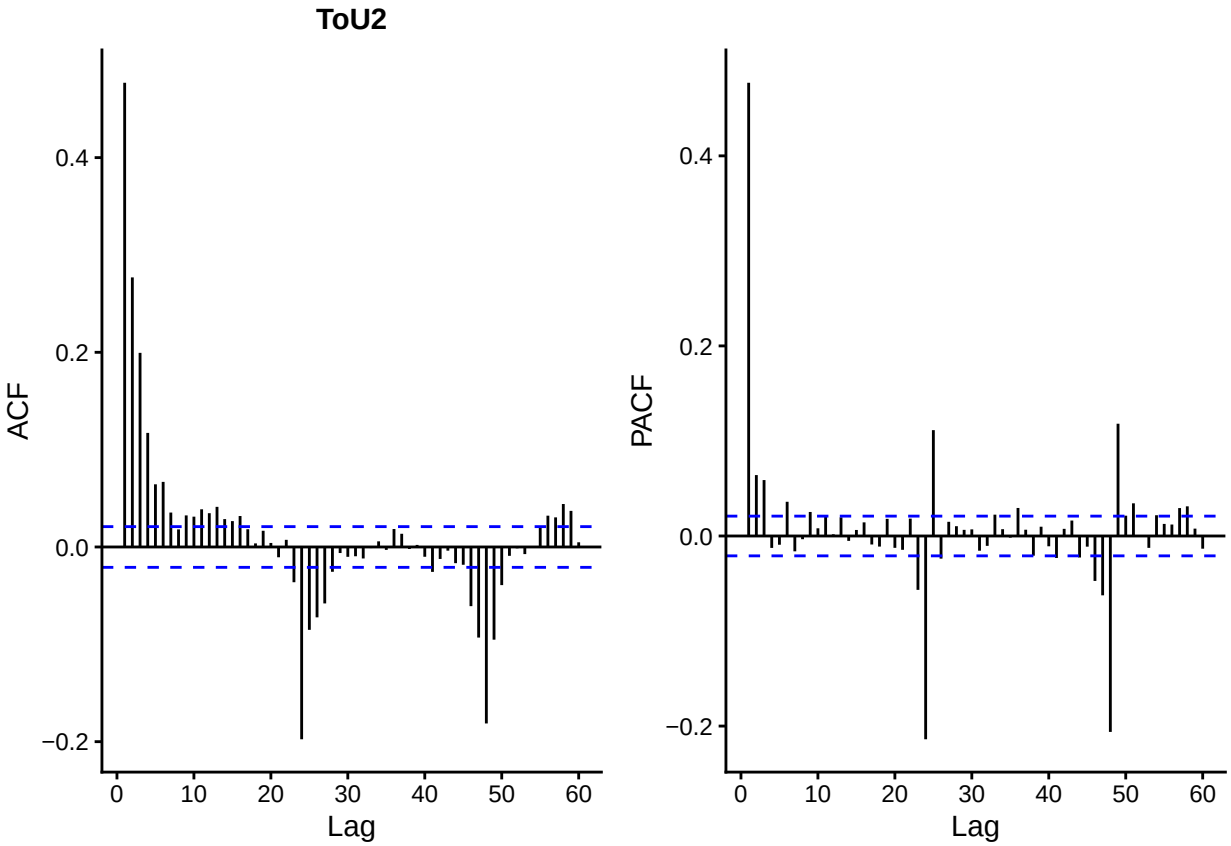

Figure 7: ToU ACF and PACF

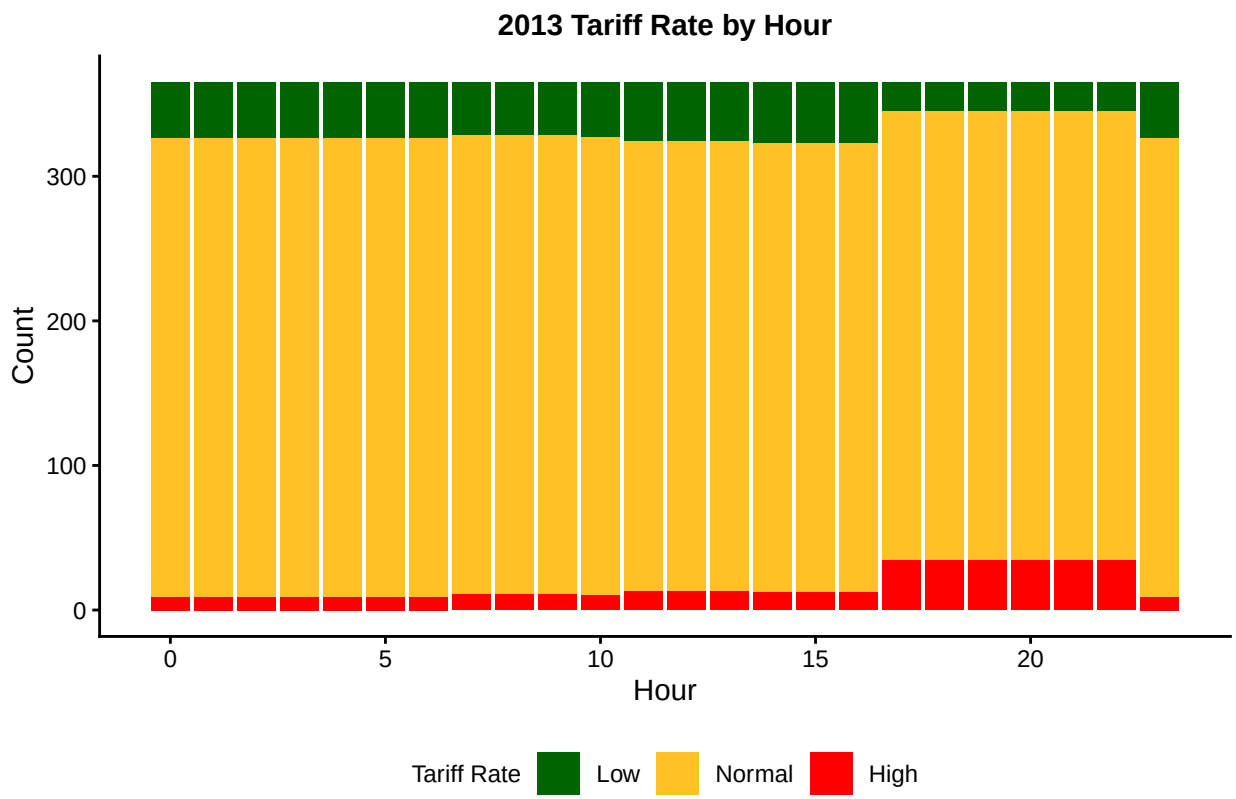

Figure XX.

Figure 8: Tariff Rate Frequency by Hour

Methods

All data clusters in this study exhibited multiple seasonalities, indicating that the best forecasting models would be advanced models capable of handling daily, weekly, monthly, and/or yearly seasonalities. Four advanced models with such capacity were selected for testing: ARIMA + Fourier terms, TBATS, NNETAR, and the Prophet model. For the ARIMA + Fourier model, the function auto.arima() was used, nnetar() for NNETAR, tbats() for TBATS, and prophet() for the Prophet model. For models with variables that could be manually adjusted (ARIMA + Fourier and NNETAR), a train and test approach was used to determine the best fitting variables on the 2012 data. The first 11 months of hourly data were trained and the last month of hourly data was used as the test case. The variable values, K for ARIMA and K, p, and P for NNETAR, that produced the lowest AIC scores for each forecast method were selected for the fitted model. The fitted model for both of these methods was then used to forecast the load data for 2013.

To forecast the models and predict 2013 hourly load for all clustered groups, the functions forecast() and predict() (Prophet only) were employed. Both the Prophet and TBATS forecasts failed to produce a reasonable 2013 load forecast and were eliminated from model selection. The TBATS model produced a straight line forecast, and the Prophet model produced a linearly growing forecast that did not capture monthly or annual seasonalities. The NNETAR and ARIMA + Fourier models both produced forecasts for 2013 that generally captured the trend and seasonality of the 2013 year.

Because the Std clusters (1 and 2) had no changes in rate structure from 2012 to 2013, Mean Absolute Percentage Error (MAPE) scores for the best fitting model are expected to be lowest in these clusters. However, the MAPE scores for each forecast all exceeded 40% indicating low accuracy in both forecasting models. Regardless, the forecast from the ARIMA + Fourier analysis model was the best fit for the hourly load data and selected as the study model due to having lower MAPE scores than the NNETAR forecasts. The final MAPE scores comparing cluster forecast to actual 2013 load data were as follows: Std1= 76.7%, Std2= 41.8%, ToU1= 49.9%, and ToU2= 45.7%.

Models

Results

The comparison of the ARIMA forecast for each of the clusters compared to the actual electricity demand in 2013 ([[**Figure 9**]]) shows a difference between standard and time of use cluster demand behaviors in response to tariff rates. In the two standard clusters, the forecast and actual are fairly similar, with their difference centered around 0 and uniform across all hours. In both time of use clusters, the forecast seems to overestimate demand during periods of "High" tariffs and underestimate demand during periods of "Low" tariffs. In [[**Figure 9**]] the time of use clusters' red curves, which represents all "High" tariff hours, are shifted to the right, indicating that the forecast value at a given hour tends to be higher than the actual value. The green curves are shifted to the left, indicating that the forecast value at a given hour tends to be lower than the actual value. This aligns with the hypothesis that customers will change their behavior and shift their load from periods of high tariff rates to low tariff rates. Notably, the Std1 cluster has a much larger residual range than any of the other clusters. Further statistical analysis should be done to determine if this visual trend is significant. Future research should investigate if overall usage in time of use households changes with the tariff implementation and whether certain patterns in hourly usage or the duration of high and low tariff periods are associated with greater behavioral change.

Summary and Conclusions

This study separated London electricity users into clusters based on participation in a ToU tariff scheme that began in 2013. Models were tested and trained via forecasting during the year prior to the tariff scheme being installed in order to generate forecasts for the 2013 year. The goal of this study was to determine if there were any behavioral changes that could be observed from forecasts of expected load demand vs actual load after the tariff 'treatment' had been applied to only the select clusters of individuals that opted in. This analysis did reveal a change in behavior that shifts demand from peak hours to off-peak hours. Load forecasts

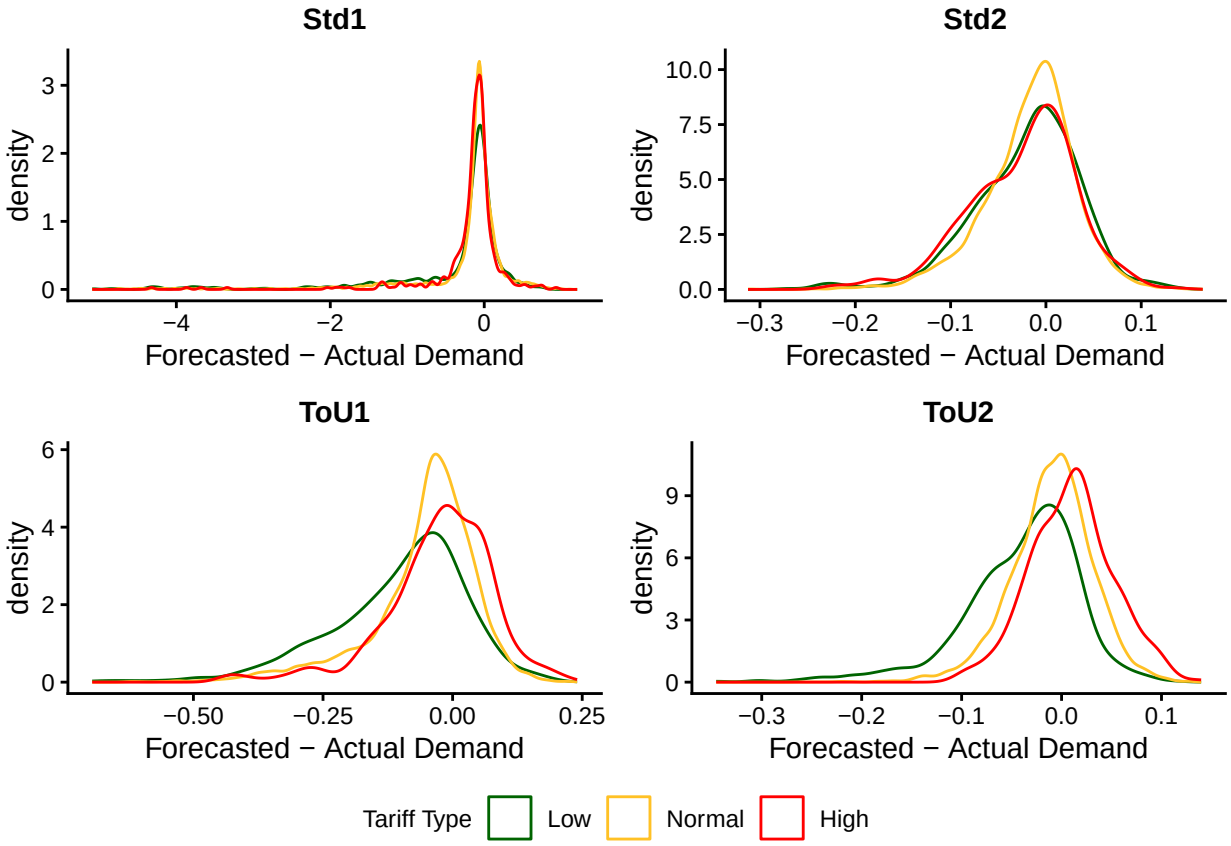

Figure 9: Forecasted Versus Actual Demand for Standard and Time-of-Use Households

from clusters of ToU tariff users showed over-prediction during peak hours for users and under-prediction during off-peak hours in the year the tariffs were implemented. Standard user clusters that did not partake in the ToU tariff scheme did not have any visible behavior shifts from expected based on the forecast.

Key limitations to this study include low model accuracy (determined from MAPE score) and lack of longitudinal data. Only one year of data, 2012, was available in sufficient quantity to cluster and perform exploratory analysis from this data-set. Similar studies might achieve greater model accuracy where it is most expected (in standard groups) if multiple years of data are available prior to the forecast year. The annual peaks and lows that the forecasting models in this study struggled to capture may be better represented by additional years for model training. Additionally, the inclusion of weather data like hourly temperature may also be utilized in future studies as an exogenous variable and aid in more accurate forecasts of the most extreme load peaks. With more accurate forecasts, the strength of conclusions regarding behavior shifts would be higher. Future research may seek to explore similar data-sets in other urban locations to compare trends spatially or across different tariff structures.

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